Response to the National Science Foundation's Request for Information on the Development of an Artificial Intelligence (AI) Action Plan

Submitted by: The Institute for Trustworthy AI in Law and Society (TRAILS)

Overview

America dominates global markets for AI. As Vice President Vance noted, "The U.S. possesses all components across the full AI stack, including advanced semiconductor design, frontier algorithms, and, of course, transformational applications."[1] In addition, the US has abundant funds for capital investment, a robust public and private research sector, extensive AI expertise, large pools of various types of data, and tools to protect AI stakeholders from harmful practices or misuse. Despite these advantages, AI will not be adopted if people do not believe that they can trust its behavior. To maintain and realize the full potential of America's AI dominance it is essential that we invest in mechanisms – across AI design, development, deployment, and governance – to ensure that the technology is trustworthy and that people know when to trust it. By fostering trust in AI, we can harness its power to drive human flourishing, ensuring that AI serves to enhance well-being, opportunity, and progress for all.

Sustaining U.S. leadership in AI demands more than continued investment or technological advantage alone—it requires trust. While other nations actively pursue strategies to enhance their AI capabilities and competitiveness, the U.S. must also distinguish its technology through reliability and trustworthiness. Recognizing this, the Institute for NSF-NIST Trustworthy AI Institute for Law & Society (TRAILS) is the first organization to integrate AI participation, technology and governance during the design, development, deployment and oversight of AI systems. We investigate what trust in AI looks like, how to create technical AI solutions that

[1] https://www.presidency.ucsb.edu/documents/remarks-the-vice-president-the-artificial-intelligence-action-summit-paris-france

build trust, and which policy models are effective in sustaining trust.[2] With that perspective in mind, **TRAILS maintains that the AI Action Plan must center on ensuring that US made AI is trustworthy**. In the pages that follow, TRAILS outlines why trust is important to the success of US AI. We also recommend several policies that the USG should take to sustain its competitiveness in AI.

All of these recommendations are situated from the position that the U.S. cannot take its technological leadership in AI for granted. Policymakers in many other countries are determined to nurture their own domestic suppliers of AI because they believe AI is both a "general-purpose technology" essential to their nations' economic growth and a "dual-use technology" essential to national security.[3] Furthermore, AI scaling laws, the methods and expectations that labs have used to increase the capabilities of their models for the last five years, are now showing signs of diminishing returns.[4] AI developers are devising new strategies and business models to create AI. For example, the Chinese company Deep Seek claims to have produced a new AI model relying on less compute, energy, and capital costs than many US or European foundation models. Finally, other governments, including China, the UAE, and Saudi Arabia have clear plans and funds to nurture and finance AI and their civil and military sectors often collaborate to fund, develop and test AI.[5]

Trust as a Foundation for AI Leadership

AI can be opaque, complex, and unpredictable.[6] To ensure that AI realizes its full potential, creators and deployers of AI must first find ways to engender trust or make AI worthy of our

[2] https://www.trails.umd.edu/

[3] S. Aaronson, AI Nationalism and its Effects, CIGI Paper No. 306, September 2024, https://www.cigionline.org/publications/the-age-of-ai-nationalism-and-its-effects/

[4] https://techcrunch.com/2024/11/20/ai-scaling-laws-are-showing-diminishing-returns-forcing-ai-labs-to-change-course/

[5] S. Aaronson, AI Nationalism,; https://www.lawfaremedia.org/article/sovereign-ai-in-a-hybrid-world--national-strategies-and-policy-responses; and https://www.brookings.edu/articles/the-global-ai-race-will-us-innovation-lead-or-lag/

[6] D. Acemoglu, Harms of AI, NBER Working Paper 29247, 2021, https://www.nber.org/papers/w2924 7

trust.[7] That alone is not enough – they must also rigorously evaluate and accurately report when AI systems are or are not trustworthy.[8] As the Executive Order on Promoting the Use of Trustworthy Artificial Intelligence in the Federal Government (December 3, 2020) states:

> "The ongoing adoption and acceptance of AI will depend significantly on public trust. Agencies must therefore design, develop, acquire, and use AI in a manner that fosters public trust and confidence while protecting privacy, civil rights, civil liberties, and American values, consistent with applicable law."[9]

Unfortunately, there is mounting evidence that trust in US AI is declining in the US and internationally. A 2024 Gallup poll found that some 50% of Americans see more harm than good from societal use of AI. Moreover, Americans don't trust that firms will use AI responsibly. But some 57% believe that greater transparency into how firms use AI could reassure them that AI is worthy of their trust.[10] A 2024 Pew Poll found similar results: 52% of Americans are more concerned than excited about AI in daily life. In contrast, only two years earlier in 2022, Pew found that only 38% were more concerned. [11] In another 2024 survey of global attitudes towards AI, IPSOS found that, although more people are using AI, they remain concerned about ensuring AI's responsible development.[12] Distrust in US AI is also rising among policymakers. Some governments now believe it could be risky to depend on other countries for access to AI or its

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[7] S. Afroogh et al. Trust in AI: progress, challenges, and future directions. Humanit Soc Sci Commun 11, 1568 (2024). https://doi.org/10.1057/s41599-024-04044-8/

[8] B. Stanton and T. Jensen, Trust and Artificial Intelligence, (Draft) March 2, 2021, https://www.nist.gov/publications/trust-and-artificial-intelligence-draft

[9] Executive Office of the President, Promoting the Use of Trustworthy Artificial Intelligence in the Federal Government, President Donald J. Trump, Executive Order 13960 of December 3, 2020, https://www.federalregister.gov/documents/2020/12/08/2020-27065/promoting-the-use-of-trustworthy-artificial-intelligence-in-the-federal-government

[10] https://news.gallup.com/poll/648953/americans-express-real-concerns-artificial-intelligence.aspx#:~:text=In%20terms%20of%20trust%2C%20in%202024%2C%20about,2024)%20trust%20businesses%20a%20lot%20or%20some

[11] https://www.pewresearch.org/short-reads/2023/11/21/what-the-data-says-about-americans-views-of-artificial-intelligence/

[12] https://www.ipsos.com/en-us/google-ipsos-multi-country-ai-survey-2025. The survey was conducted between September 17th- October 8th, 2024, on behalf of Google. For this survey, a sample of 21,043 adults age 18+ were interviewed online in 21 countries.

components.[13] For example, in 2024, Dutch officials expressed concerns that they are creating a national security risk by relying on US firms for AI infrastructure.[14] Australian officials expressed similar concerns.[15]

The sections below delineate TRAILS' recommendations on how the US Government can encourage the development of AI that is trustworthy, thereby increasing the chances of successful adoption of this technology. We also argue that a well-informed, whole-of-society public debate is necessary for capturing the beneficial potential of the technology, while limiting the risks associated with it.

Hence, in the Action plan, the US should:

Invest in Research to Promote Trustworthy AI

AI is advancing at an unprecedented pace, but trust remains a critical barrier to its widespread adoption. Private-sector AI development often focuses on speed to market, which can leave gaps that degrade trust: in transparency, oversight, and consistent evaluation standards, among others. Without robust mechanisms to assess AI's trustworthiness, the U.S. risks falling behind in ensuring that AI systems are safe, reliable, and widely accepted. This, in turn, gives an opportunity for foreign alternatives to become more widely adopted. Public investment in AI research can bridge these gaps by supporting rigorous, evidence-based frameworks that ensure AI is transparent, accountable, and aligned with human needs. Collaborative efforts—such as the partnership between TRAILS and the National Institute for Standards and Technology (NIST) — are already paving the way for stronger evaluation methods and shared benchmarks. Expanding these efforts will not only accelerate AI innovation but also strengthen public trust and global

[13] M Alduhishy, Sovereign AI: What it is, and 6 strategic pillars for achieving it, World Economic Forum, April 25, 2024, https://www.weforum.org/stories/2024/04/sovereign-ai-what-is-ways-states-building/

[14] Government of the Netherlands. 2024. The government-wide vision on Generative AI of the Netherlands. The Hague, The Netherlands: Ministry of the Interior and Kingdom Relations. January 17. 2024, p. 14, www.government.nl/documents/parliamentary-documents/2024/01/17/government-wide-vision-on-generative-ai-of-the-netherlands.

[15] G. Bell, et al, . Rapid Response Information Report: Generative AI: Language models and multimodal foundation models. Australia's Chief Scientist. March 24, 2023,www.chiefscientist.gov.au/GenerativeAI

confidence in U.S.-developed AI technologies. To achieve this, the government should prioritize research in areas that complement and enhance private-sector efforts rather than duplicating them, including:

Participatory AI

AI works best when it serves a broad range of users,[16] yet foundational AI research often proceeds without input from those affected, adversely or otherwise, by its deployment. This lack of engagement can limit AI's usefulness and adoption. To strengthen the global leadership of U.S.-developed AI, the government should invest in participatory research methods that involve businesses, workers, technical experts, consumers, and civil society at every stage of AI design and deployment. Such investments complement, rather than duplicate, private-sector efforts by ensuring AI are not only commercially viable but also widely applicable and trusted.[17] Strengthening participatory research will reinforce U.S. leadership in AI development while promoting innovation and sustainable growth in the private sector.

Technical methods and metrics to ensure trustworthy AI systems

Trust in AI must be grounded in rigorous, reliable evaluation. Private companies focus on AI innovation and performance, but their evaluation methods are often proprietary and inconsistent, making it difficult to assess reliability across the industry. Public investment in standardized evaluation, verification, and benchmarking—through initiatives like NIST's Assessing Risks and Impacts of AI (ARIA) program and the AI Safety Institute Consortium—provides independent, trusted frameworks that strengthen industry credibility, ensure fair competition, and enhance global confidence in U.S.-developed AI. Policymakers should support research that creates user-facing technologies to clarify AI behavior, transparency, privacy, and reliability. These investments also promote marketplace stability, ensuring that companies compete on services provided.

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[16] F. Delgado et al. The participatory turn in ai design: Theoretical foundations and the current state of practice. In *Proceedings of the 3rd ACM Conference on Equity and Access in Algorithms, Mechanisms, and Optimization, October 2023,* (pp. 1-23).

[17] K. Shilton. Values levers: Building ethics into design. *Science, Technology, & Human Values*, *38*(3 2013, pp.), 374-397, https://www.jstor.org/stable/23474474

Empirical evaluation of AI's judgments and decisions

Trust in AI starts with understanding it – how these systems make judgments, why they sometimes produce unexpected results, and how users interpret their outputs.[18] Currently, many AI users lack a clear understanding of how these technologies function, when they may behave unpredictably, and how to assess the reliability of their responses.[19] This uncertainty can limit adoption and confidence in AI-driven solutions. To address this, scientific research should focus on developing methods and tools that clarify AI decision-making, predict when errors are likely to occur, and improve evaluation processes. Industry and regulatory bodies also need consistent, reliable approaches to assess AI systems against established standards while ensuring that oversight does not hinder progress. By systematically analyzing how AI aligns with real-world expectations and addressing instances where it replicates flawed human reasoning, the U.S. can strengthen reliability, confidence in AI systems, and long-term global competitiveness.[20] Investing in structured evaluation methods will help ensure that AI not only advances technologically but does so in a way that is measurable, verifiable, and dependable across a range of applications.

Research-driven AI governance

Strong AI governance starts with research, not guesswork. Standardized governance frameworks, developed through empirical research, can enhance safety, fairness, and interoperability while avoiding unnecessary regulatory burdens that might hinder innovation. Private AI companies often create internal standards without regulator or expert consultation, leading to fragmented approaches to AI safety, bias mitigation, and transparency. Independent research institutions help bridge this gap by developing widely accepted governance frameworks, with organizations like NIST, ISO, and IEEE playing a key role in drafting AI guidelines that emphasize transparency

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[18]D. Broniatowski *Psychological foundations of explainability and interpretability in artificial intelligence* (Vol. 4, 2021, p. 00). US Department of Commerce, National Institute of Standards and Technology, https://www.nist.gov/publications/psychological-foundations-explainability-and-interpretability-artificial-intelligence

[19] S. Edelson et al., How Decision Making Develops: Adolescents, Irrational Adults, and Should AI be Trusted with the Car Keys? *Policy Insights from the Behavioral and Brain Sciences*, *11*(1), 2024, pp. 11-18, https://psycnet.apa.org/record/2024-67503-002.

[20] E. Tabassi, Artificial Intelligence Risk Management Framework (AI RMF 1.0), January 26, 2023, https://www.nist.gov/publications/artificial-intelligence-risk-management-framework-ai-rmf-10

and accountability. For example, the NIST AI Risk Management Framework (AI RMF) exemplifies how research-backed models can guide responsible AI deployment, enabling businesses to compete on meaningful advancements like accuracy, explainability, and security.

To date, TRAILS has completed almost two years of research on trustworthy AI. Our recommendations to the AI Action plan build on our findings, specifically:

Provide Incentives to AI Firms to Encourage them to Reveal Information about Data Provenance

The data sets for foundation models are large, diverse and multinational, and are thus difficult to govern. But the world must do more to govern these models for two reasons: first, because many of these systems are black boxes, whose developers provide little information about how they work; and second, because more and more people rely on these models for information. Policy makers should aim to ensure that the data sets that underpin these models are accurate, complete and representative. Recent research indicates that increased dataset transparency can enhance interoperability, improve data quality, and encourage responsible data stewardship. Efforts like Datasheets for Datasets[21] and the NIST AI RMF 2023[22] highlight how provenance tracking can strengthen AI accountability by enabling authenticity verification and in so doing, mitigating downstream risks. Requiring AI developers to undergo audits by trained outside experts on foundation models and data—ensuring provenance, testing methods, and data usage—would allow auditors to verify that firms provide complete and accurate information.

Support International Research Collaborations as a Means of Building Trust

According to Statista, the US will remain the largest market for AI. But AI growth markets are overseas in middle income and developing countries such as Indonesia and India where the

[21] T. Gebru et al. Datasheets for Datasets CACM, December 2021, https://arxiv.org/abs/1803.09010

[22] E. Tabassi, Artificial Intelligence, and the 2024 Generative AI Risk Management Framework, https://www.nist.gov/itl/ai-risk-management-framework

population is growing.[23] For this reason, the US should work with our allies on shared approaches to research, data governance, and AI Safety. NSF has developed programs with other governments including the Quad Nations (India, Japan, Australia), Israel, and the UK. Such shared funding not only yields new research, but can build shared norms, relationships and trust between AI stakeholders in these nations.[24]

Use the Evidence-Based Policy Making Act as a Model for Developing AI Policies

In 2018, President Trump signed the Foundations for Evidence-Based Policymaking Act which encouraged collaboration and coordination to advance data and evidence-building functions in the USG. The act required that federal agencies appoint evaluation, statistical, and chief data officers and develop an evidence-building plan every four years as well as an annual evaluation plan. The USG should build on this legislation in designing the AI Action plan as well as annual evaluation plans.

Encourage Greater Understanding of AI and Greater Participation in AI Governance

The US has long used the Federal Register to solicit public comment on governance, as it is doing with this call. But research has shown that the US does not make a broad effort to market such calls from public comment. Consequently, officials often receive only several hundred comments, primarily from the technical community or businesses. Policymakers are thus missing an opportunity to build trust in AI by not using the process to involve a broader cross-section of their constituents. Moreover, discussions with a broader cross-section of their citizens may enable officials to anticipate future problems related to AI and even to gain new insights into how citizens use AI. Policymakers could incentivize such participation by holding regional

[23] Statista, Artificial Intelligence – Worldwide, https://www.statista.com/outlook/tmo/artificial-intelligence/worldwide

[24] https://www.nsf.gov/oise/international-collaborations#the-quad-australia-india-japan-and-the-united-states-6f7

townhalls and online webinars where they can talk with their constituents.[25] More generally, policymakers should foster partnerships among academia, industry, and professional organizations to create research-supported governance frameworks that clearly define standards for safety, fairness, and effective integration without hindering innovation.

Conclusion: Trust as the Cornerstone of AI Leadership

The future of AI leadership in the United States hinges on building and sustaining trust—among researchers, developers, policymakers, and the public. AI systems that are transparent, accountable, safe, and aligned with societal norms will not only gain public confidence but will also enhance U.S. innovation, competitiveness, and security in an increasingly AI-driven world.

We appreciate the opportunity to provide input and look forward to continued collaboration on shaping the AI Action Plan.

Respectfully submitted,
The Institute for Trustworthy AI in Law and Society (TRAILS)

[25] S. Aaronson, A Dysfunctional Dialogue About AI-NTIA and the Public on The Risks and Benefits of Open Foundation Models December 20, 2024, https://papers.ssrn.com/sol3/papers.cfm?abstract_id=5176397 and S. Aaronson and A. Zable, Missing Persons: The Case of National AI Strategies, CIGI Paper No 283, March 2023, https://www.cigionline.org/publications/missing-persons-the-case-of-national-ai-strategies/